Median alert lead time tracks how long patients were observed, and runs opposite to specificity

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Abstract

Continuously monitored clinical prediction models are often summarized by a median alert lead time: how many minutes before an event the first alert fires. Because a first alert can only fall inside the window a patient was observed before the event, this number depends on the cohort's observation windows as well as on the model. Across three perioperative cohorts and five model families, median lead rose by 26 to 54 minutes from the shortest to the longest observed-window band in all fifteen model-by-cohort combinations. Longer leads also came with worse operating points: median lead ran opposite to case-level specificity (Spearman -0.803), and the longest median lead, 24 minutes, was read at a specificity of 0.406. Two natural repairs leave the dependence in place. TRIPOD+AI asks for neither the window distribution nor the operating point beside the lead, and none of seven published lead-time reports we examined summarizes the observation window or stratifies the lead by it. We propose four disclosures that make a reported lead time interpretable.

1 Introduction

A lead time is the number a monitoring model is sold on. A model that is read continuously during care is asked how early it warns, and the usual answer is the **median alert lead time**: the interval between the first minute a case's score crosses an alert threshold and the onset of the event, taken over the event cases the model detected. It is reported for sepsis [Wong et al., 2021], for acute kidney injury [Lee et al., 2026], and for intraoperative hypotension, where a commercial index has been evaluated on how far ahead it warns [Hatib et al., 2018, Davies et al., 2020, Wijnberge et al., 2020, Maheshwari et al., 2020]. The label is already plain language, so it travels well beyond the paper that computed it: "the system warns a median of 24 minutes ahead" reads as a property of the system, to a clinician, a purchaser or a regulator alike.

The number also measures how long patients were watched. A first crossing can only be found inside the pre-onset window over which the patient was observed, so the lead is bounded above by that window, and the window is set by when the event happens relative to the start of monitoring, not by the model. That a lead cannot exceed its window is obvious. What has not been measured is how large the resulting dependence is in realistic evaluation pipelines, and what else a long reported lead conceals. Table 1 sets each reading the label invites beside what the number supports in our data.

What we find. On three perioperative cohorts and five model families:

1. **The dependence is large**: up to 54 minutes within one model and cohort (Section 3).

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Table 1 | What the label invites, and what the number supports. Column one is the reading the label makes available, not a measured reader response. Column two is what this paper's measurements support, and column three the disclosure that would separate the two.

What the label invites	What the number supports here	The disclosure that would close the gap
The model warns about 24 minutes ahead	A median over detected event cases, itself a window-dependent set, under this cohort's window distribution (Section 3)	The pre-onset window distribution, and a window-conditional timing statistic (Section 6.3)
Twenty-four minutes is a property of the model	Within one model and cohort, median lead rises across window bands by 26 to 54 minutes, in 15 of 15 cells (Table 2)	Any single median stratified by the realized available window (Section 6.3)
Twenty-four minutes of usable warning	A lead read at case-level specificity 0.406, the lowest of the five families in that cohort (Table 2)	The operating point reported beside the lead (Section 4)
A longer lead indicates a better model	Lead runs opposite to specificity across the fifteen combinations, Spearman -0.803 (Section 4)	Lead and operating point reported together, not lead alone (Section 6.3)

2. **Longer reported leads came with worse operating points** (Section 4).
3. **Two natural repairs do not remove it** (Section 5).
4. **Neither TRIPOD+AI nor published reports provide the window**, and we propose four disclosures that would (Section 6).

Related work. Early-warning evaluation has been criticized before: the C-statistic has been argued uninformative for early warning scores, with per-alert and workload measures proposed instead [Romero-Brufau et al., 2015]; monitor alerting has been quantified as an alarm burden per patient-day [Paine et al., 2016]; the choice of evaluation sample moves reported performance for this exact task on these public data [Yang et al., 2025]; the task's retrospective traces its arc from hope to hype [Michard and Futier, 2023]; and cross-center evaluation separates learned methods from pressure-derived rules far less cleanly than single-center reports suggest [Chaari et al., 2025]. Closest in spirit, Lauritsen et al. [2021] shows discrimination and calibration moving with the developer's choice of observation, prediction and lead windows, and asks that a standard record those choices. Here the window is not a design choice but a realized, patient-varying quantity, and what moves is the timing statistic itself. This paper grows out of our own earlier work on acute kidney injury [Lee et al., 2026], which evaluated the same transformer family under simulated continuous monitoring and reported median lead times of 73 to 93 hours as a pooled summary. The present paper asks what such a summary needs beside it to be read correctly, and Table 3 scores that earlier report together with the others.

2 Setting

Task and outcome. Intraoperative hypotension is mean arterial pressure (MAP) below 65 mmHg sustained for at least three consecutive minutes. The 65 mmHg level is conventional [Salmasi et al., 2017, Sessler et al., 2019]; the three-minute sustain is this study's operationalization. Accumulated exposure is associated with kidney and myocardial injury after surgery [Walsh et al., 2013, Salmasi et al., 2017, Wesselink et al., 2018], and none of the responses acts instantly, so a warning interval is worth what a clinician can deliver inside it.

Cohorts. Models were developed on the EPIC arm of MOVER and evaluated on its held-out test split (1,404 cases, 786 with a sustained event) and on two external cohorts: VitalDB [Lee et al., 2022, Lee and Jung, 2022], with 2,804 cases and 1,007 events, and the SIS arm of MOVER [Samad et al., 2023, UCI Machine Learning Repository, 2023], with 2,641 cases and 1,409 events. The two MOVER arms come from one hospital and were de-identified separately, so patient overlap between them cannot be measured; VitalDB, from a different country and health system, has no such question. All intervals are 95% cluster bootstrap, over patients, or over cases in MOVER (SIS), which releases no usable subject key.

Models. Five families: a transformer over irregularly sampled (type, time, value) triplets [Tipirneni and Reddy, 2022], the ITE (initial triplet embedding) Transformer; an LSTM; XGBoost; random forest; and logistic regression. Appendix A.1 gives architectures, training and compute. No model is proposed for clinical use, and nothing here is a ranking of architectures.

Table 2 | Lead time, its window dependence, and the operating point it was read at. One row per model and cohort, at the transported 80%-sensitivity threshold. `Band lead` is the median lead, in minutes, in the shortest and the longest band of available pre-onset window, and `Rise` their difference (intervals, Table A1). `Pooled lead` is the single median, in minutes, a paper would report. Sensitivity and specificity are case-level (Section 2; intervals, Table A6). `Case AUROC` is the threshold-free discrimination of the case's maximum score over its alert window, the statistic the alert rule thresholds, with its 95% cluster bootstrap interval.

Model	Cohort	Band lead	Rise	Pooled lead	Sens.	Spec.	Case AUROC (95% CI)
Transformer	MOVER (EPIC)	9 → 46	+37	24	0.777	0.406	0.651 [0.621, 0.677]
Transformer	VitalDB	5 → 59	+54	17	0.679	0.379	0.642 [0.619, 0.665]
Transformer	MOVER (SIS)	6 → 46	+40	22	0.657	0.467	0.641 [0.621, 0.662]
LSTM	MOVER (EPIC)	6 → 44	+38	23	0.807	0.474	0.693 [0.667, 0.719]
LSTM	VitalDB	3 → 34	+31	9	0.492	0.812	0.771 [0.754, 0.787]
LSTM	MOVER (SIS)	5 → 31	+26	15	0.551	0.709	0.716 [0.694, 0.736]
XGBoost	MOVER (EPIC)	8 → 38	+30	21	0.810	0.608	0.766 [0.741, 0.791]
XGBoost	VitalDB	3 → 30	+27	13	0.694	0.754	0.807 [0.792, 0.822]
XGBoost	MOVER (SIS)	5 → 36	+31	19	0.790	0.638	0.773 [0.754, 0.792]
Random forest	MOVER (EPIC)	8 → 38	+30	22	0.828	0.555	0.746 [0.717, 0.772]
Random forest	VitalDB	3 → 34	+31	15	0.803	0.674	0.793 [0.777, 0.809]
Random forest	MOVER (SIS)	6 → 40	+34	22	0.849	0.559	0.764 [0.746, 0.781]
Logistic reg.	MOVER (EPIC)	8 → 40	+32	23	0.837	0.518	0.718 [0.690, 0.746]
Logistic reg.	VitalDB	3 → 28	+26	14	0.461	0.825	0.776 [0.760, 0.793]
Logistic reg.	MOVER (SIS)	5 → 38	+33	21	0.790	0.585	0.723 [0.703, 0.743]

Deployment simulation and operating point. Every model is read forward at every eligible minute of every case, using only measurements recorded before that minute and after a source-specific induction exclusion. A case **alerts** if its score reaches the threshold at any minute in the 60 minutes before the event, and its **lead time** is the interval from the first such minute to the event. **Case-level sensitivity** is the share of event cases that alert and **case-level specificity** the share of control cases that never do. Thresholds were fitted on the development validation split at an 80% case-level sensitivity target and **transported** unchanged to every cohort; nothing is reselected, retrained or re-scored.

The window the rule searches. A case's **available pre-onset window** is its observable run-up to onset after the induction exclusion, capped at the 60-minute alert window. A short window is not a poorly monitored case: the event arrived early in the record, so there is no missing data to recover. An alert already active at a case's first eligible minute is **prevalent**. Control cases take a pseudo-onset sampled from the event cases' timing distribution, so they are scored over a comparable window.

3 The reported lead moves with the observed window by up to 54 minutes

Median lead rises with window length in all fifteen model-by-cohort cells, by 26 to 54 minutes (Table 2). Splitting detected event cases into quantile-derived bands of available window, the rise from the shortest to the longest band has a pointwise interval excluding zero in **15 of 15** cells, and amounts to **0.51 to 1.02 minutes of lead per minute of window difference** (Table A1). For the transformer the shortest-window band reports 5 to 9 minutes and the longest 46 to 59; its pooled median of 17 to 24 minutes sits between them because that is where the median case happens to fall (Figure 1, Table A2). A pooled median is therefore as much a description of the cohort's window distribution as of the model.

The same fact as a ratio. The median ratio of a case's lead to its own window is 0.455 to 0.944, and the Spearman correlation between lead and window is 0.501 to 0.906, across the fifteen cells; the transformer is the extreme at 0.899 to 0.944. Detection also depends on window length (Table A3), so for the tabular families part of the rise may be survivorship.

How much of this exceeds the ceiling, we cannot say, and the reporting point does not depend on it. Because a lead cannot exceed its window, some positive dependence arises by construction. A reference that places a pseudo-crossing uniformly on each case's window has median ratio 0.518 to 0.534, which the transformer's observed ratio exceeds by **+0.372 to +0.421**; in **2 of 15** cells,

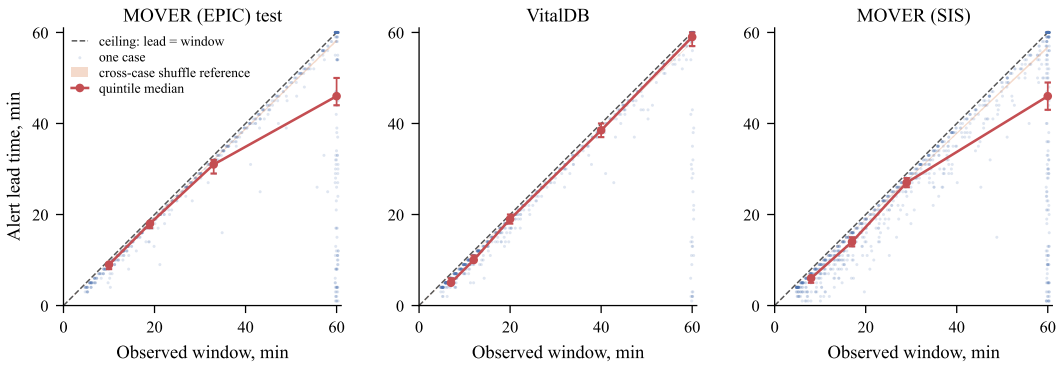


Figure 1: **Alert lead time against the available pre-onset window.** Every detected event case under the ITE Transformer, three held-out cohorts; the five-family version is Figure 2. The dashed diagonal is the ceiling. The shaded band is the 2.5th to 97.5th percentile envelope of the cross-case shuffle reference, not a per-case prediction interval. Red markers are the band medians of Table A2 with 95% cluster bootstrap intervals.

all VitalDB, the observed ratio sits *below* it. A cross-case shuffle that gives each case another case's score sequence reproduces the association: its median ratio is **0.72 to 0.97** and its Spearman **0.63 to 0.94**, at or above the observed value in **15 of 15** cells, and a length-matched variant is **not distinguishable from the observed correlation in 9 of 15 cells** (Table A4). Both shuffles are biased toward agreement, so they bound nothing; we do not claim a patient-specific fraction. For reporting this does not matter: whatever its source, a pooled median mixes cases whose leads differ by tens of minutes because their windows did.

4 Longer reported leads came with worse operating points

Across the fifteen cells, median lead runs opposite to case-level specificity (Spearman -0.803; 86 of 99 untied pairs) and to PPV in excess of prevalence (-0.719), and the ordering holds within each cohort (9 of 9, 8 of 10 and 9 of 9 untied pairs; Spearman -0.97, -0.70 and -0.97). This is descriptive: the cells share cohorts and fits, and transported thresholds give realized sensitivities from 0.461 to 0.849 (Table 2). But it is enough for reporting: **a reader ranking these systems by their published lead would be ranking them against their operating points.**

The transformer has the longest leads, the lowest specificity and the lowest case-level AUROC in every cohort. Its median leads, 24, 17 and 22 minutes, are the longest of the five families in two cohorts and tied for longest in the third. Its specificity is 0.406, 0.379 and 0.467, so it alerts on 53 to 62% of control cases, and its PPV sits 0.065, 0.021 and 0.051 above prevalence. Its case-level AUROC is 0.651, 0.642 and 0.641, against 0.693 to 0.807 for the other four families (Table 2), so the low specificity reflects a weak case-level discriminator and not only a badly placed threshold. A reader given only "24 minutes" sees none of this.

The mechanism is simple: its score is above threshold almost from the first minute. Within three minutes of its first scored minute the transformer's median score clears its own threshold in every cohort, in event and control cases alike; in MOVER (EPIC) it rises from 0.530 to 0.938 in one minute in control cases and from 0.541 to 0.951 in event cases, against a threshold of **0.894** (Figure A1, Appendix A.7). A case whose score settles above threshold that early registers its first crossing as soon as its alert window opens, so its lead is whatever that window holds. **20.0 to 52.3% of the transformer's event minutes and 2.5 to 18.3% of its control minutes** sit at or above threshold, against 0 to 26.7% and 0 to 1.7% for the other four families.

The window dependence does not need a degenerate model. XGBoost and the random forest have specificity 0.555 to 0.754 and case-level AUROC 0.746 to 0.807, usable operating points by the standard of this task, and their median leads still rise by 27 to 34 minutes across window bands (Table 2, Figure 2). The transformer makes the specificity cost visible; the window dependence is present in every family.

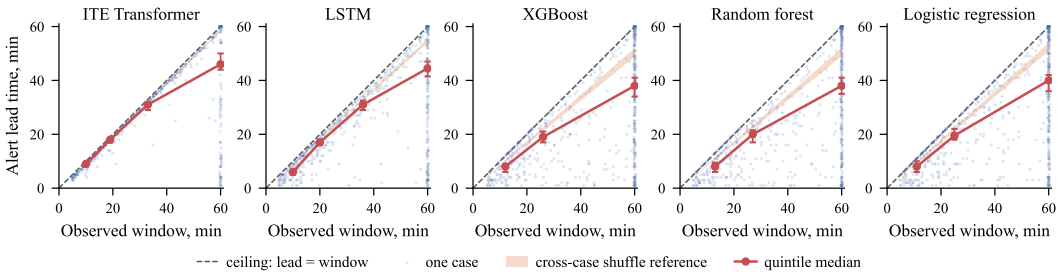


Figure 2: **Alert lead time against observed window length, five model families, MOVER (EPIC) test.** Axes and overlays as in Figure 1. The lead climbs the diagonal in every family, including XGBoost and the random forest, whose operating points are usable; the transformer is the limiting case rather than the exception.

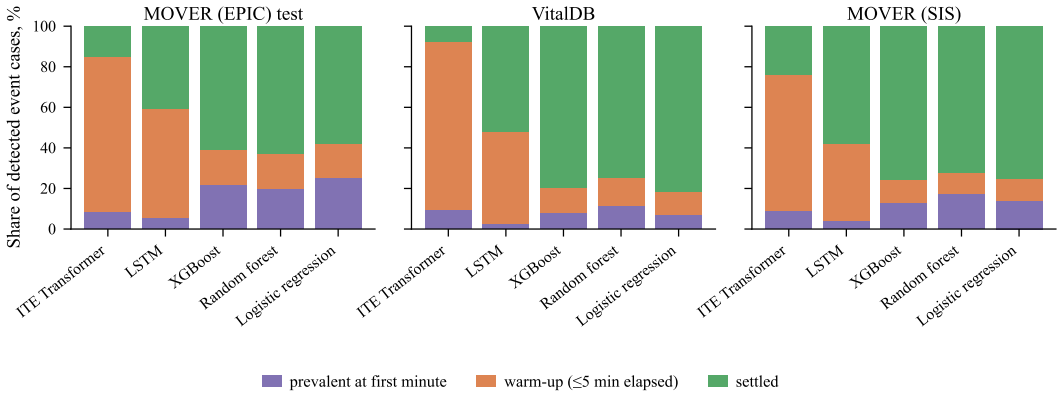


Figure 3: **Composition of detected event cases by elapsed monitoring time at the first crossing.** Prevalent cases are above threshold at their first eligible minute; warm-up cases cross within five minutes of it; settled cases cross later. Five model families, three cohorts, at the transported 80%-sensitivity operating point. The transformer's alerts are concentrated in the first five minutes, the LSTM's partly so and the tabular families' not, while the window dependence of Section 3 holds in all of them. This is the stacked bar Section 6.3's first item asks for.

None of this is a statement about physiology. MAP is both an input channel and the quantity the label is computed from, so nothing here shows that a score foresees a physiological process. Under a pre-specified rule the score does not depart from its own baseline before MAP departs from its (Appendix A.12).

5 Two natural repairs, and why neither works

Prevalent alerts are not the explanation. Alerts already active at the first eligible minute are **2.6 to 25.5%** of detected event cases across the fifteen cells, and recomputing the median lead on the remaining alerts alone costs **0 to 3 minutes in all fifteen cells** (Figure 3, Appendix A.13). The window dependence is carried by ordinary cases. Prevalence stays low because most detected event cases have windows shorter than 60 minutes that open while the score is still settling: a prevalent alert and a warm-up crossing are the same behavior either side of the first eligible minute, which is why Figure 3 shows both.

Suppressing the first minutes of monitoring is a poor trade. Discarding every minute at or before $N = 5$ minutes of elapsed monitoring and applying the unchanged threshold costs **3.6 to 6.8 percentage points of case-level sensitivity** for **0 to 3.2 percentage points of specificity**, and moves median lead by $-$**3 to +1 minutes**, in two cells *increasing* it because short-window cases drop out of the detected set. The surviving alerts still fire at the first opportunity within whatever window

remains. Lowering the threshold to restore sensitivity cannot recover more specificity than the fixed-threshold gain, since suppression only removes alerting minutes. **We do not recommend the rule**: the problem is not noisy early minutes but the coupling of the reported quantity to the window it is measured in.

Window-conditional statistics reduce the dependence, at a cost. Both repairs above change the alert rule and keep the estimand; the alternative is to change the estimand, as landmarking does [Van Houwelingen, 2007]. A **horizon-specific detection rate**, over event cases observed for at least X minutes before onset, runs 0.278 to 0.692 at a 10-minute horizon and 0.173 to 0.547 at 30 minutes, a smooth decline a single median hides (Table A8); cases observed longer than X still contribute unequal alerting opportunity. **Restricting to cases observed for the whole 60-minute window** makes the window constant: the transformer then reports the longest median lead of the five families in all three cohorts, 47, 60 and 57 minutes, while detecting only 0.441, 0.247 and 0.304 of those cases against 0.762 to 0.910 for XGBoost and the random forest (Table A9). The restriction discards 53 to 60% of event cases, the short-window cases a deployed monitor will meet, and those cases differ in surgery duration and case mix (Appendix A.16). Neither statistic repairs the pooled median; both make visible what it hides.

6 What TRIPOD+AI asks for, and four disclosures

A clinician, a regulator or a procurement reviewer meets a reported lead as a number and a name, not as a scoring policy, so what a reporting standard requires alongside it limits what any of them can conclude. We therefore ask what TRIPOD+AI [Collins et al., 2024] guarantees a reader receives next to a lead time, and whether published reports supply the rest.

6.1 The standard asks for the ingredients, not the pairing

The authors completed an item-level TRIPOD+AI review for this work before the analyses of Sections 3 to 5 were run. A lead time is declared under **item 12e**, performance measures and rationale; **item 15** covers threshold selection and **item 23a** performance with uncertainty, but none asks for the lead to be reported with its operating point. **Items 20a and 20b come closest**: they request follow-up time as a participant characteristic, which this study answers with surgery duration and first-onset time. What no item requests is the distribution of the window the alert rule searches, although that window correlates with the reported lead at $\rho = 0.501$ to 0.906. **Item 14, model fairness**, is a near miss of another kind: our answer already disclosed that arterial-line placement selects longer, higher-risk operations, but as a limitation of generalizability, unconnected to the timing metric. Table A10 lists the remaining items. TRIPOD+AI is a general prediction-model standard, not one for alerting rules, and we checked it alone; DECIDE-AI and TRIPOD-LLM were not reviewed.

6.2 Published reports give the lead without its window

None of seven published lead-time reports we examined summarizes the window over which the first alert was searched, or stratifies the lead by it (Table 3). Most give only length of stay as a cohort descriptor; Henry et al. [2015] shows each patient's window as a bar in a figure, truncated at 120 hours, without a summary. The operating point fares better: two report sensitivity and specificity in the same sentence or table row as the lead, Churpek et al. [2020] for every threshold, which is the pairing Section 6.3 asks for; Wong et al. [2021], an external validation of a widely deployed proprietary model, gives them in the preceding sentence at the same threshold; and Koyner et al. [2018] in the adjacent sentence, for a different prediction target. Our own earlier report is among those that omit the window, though its cohort required at least 72 hours of pre-event data. These are reports we had collected, not a systematic sample, so they show that the omission occurs in practice, not how often. Fixed-horizon designs, such as the derivation of the hypotension prediction index [Hatib et al., 2018, Davies et al., 2020], are not first-crossing statistics and are not scored.

6.3 Four disclosures for continuously monitored models

Each is tied to a measurement in this paper and computable from artifacts a study of this kind already produces; none requires a new experiment.

Table 3 | Seven published reports of an alert lead time, scored against Section 6.3. `Window` is whether pre-event observation time is summarized or the lead stratified by it; `Operating point` is whether sensitivity and specificity (or PPV) appear beside the lead. Values are as printed in each paper; h, hours. Five reports time the lead from the first alert or identification; Koyner et al. [2018] and Giannini et al. [2019] do not state which alert they time.

Report	Task	Reported lead, median (IQR)	Window	Operating point beside lead
Henry et al. [2015]	Septic shock	28.2 h (10.6–94.2)	Per-patient bars in a figure, truncated at 120 h; no summary	Yes, same sentence
Kang et al. [2016]	Ward deterioration	24.7 h and 29.6 h	No; length of stay only	Threshold only; specificity elsewhere
Koyner et al. [2018]	Acute kidney injury	41 h (12–141)	No; length of stay only	Adjacent sentence, for a 24-hour target
Giannini et al. [2019]	Severe sepsis	6 h 34 min (0 h 50 min–53 h 19 min)	No	Elsewhere, from a separate cohort
Churpek et al. [2020]	Acute kidney injury	27, 34.5 and 39 h, three sites	Supplementary cumulative-incidence figure, not linked to the lead	Yes, same table row
Wong et al. [2021]	Sepsis, proprietary vendor model	2.5 h (0.5–15.6)	No	Preceding sentence, same threshold
Lee et al. [2026]	Acute kidney injury	84, 93 and 73 h, three cohorts	No; length of stay only	Separate tables at that threshold; detection and alert burden at another

1. **Alert composition by elapsed observation time**: for every operating point, the prevalent, warm-up and settled shares of alerts, as in Figure 3. One stacked bar would have made Section 4 visible on sight.
2. **A window-conditional timing statistic, with the window distribution**: a horizon-specific detection curve or a fixed-window estimand as the primary timing quantity, with its denominator stated, and any single median stratified by available pre-onset window (Section 3, Tables A8 and A9).
3. **The operating point beside the lead, and N(t) beside any event-aligned curve.** Specificity or PPV belongs in the same row as the lead (Table 2). For trajectories, the risk set here roughly halves between the event and 60 minutes before it (1,404 to 708, 2,804 to 1,433 and 2,641 to 1,374), and recomputing the same bins on only the cases observable back to 60 minutes moves the first bin reaching AUROC 0.70 from 15 minutes in all three cohorts to 20, 35 and 20 (Figure A3): the horizon a reader would quote moves by up to twenty minutes on a reporting choice alone.
4. **A path-versus-displacement summary for any score read as a trend**, so that a series traveling much farther than its net displacement is not described as rising (Table A5, Appendix A.9). This is closer to a diagnostic than a disclosure, and the weakest of the four.

Four items are a burden, so they should replace text rather than add to it. Each is one figure or one row beside a number a paper already reports; if there is room for only one, it should be the operating point in the same row as the lead. The magnitudes here will not transfer to other settings, but the structure will: any continuously read model summarized by a first-crossing statistic over patient-varying windows is susceptible. A median lead should be reported beside its window distribution and its operating point, never alone.

7 Limitations

Retrospective, one label, one fit. Every score comes from a retrospective forward-running simulation on cached data, not deployment. The outcome's level and duration are conventions we did not vary. Models were trained at one seed and thresholds come from one validation split at one sensitivity target; a different fit or target could move the magnitudes in Sections 3 to 5. Control cases have no onset, so each takes one pseudo-onset drawn at a fixed seed, and every control-side quantity

(specificity, PPV, the association of Section 4) is conditional on that draw; its variability is not in any interval we report.

Two cohorts share a hospital. MOVER's EPIC and SIS arms come from the same site and were de-identified separately, so patients may appear in both the development cohort and MOVER (SIS), and that overlap cannot be measured. VitalDB, from another country and health system, is the external cohort without this caveat, and both main results, the window dependence and the lead-specificity ordering, hold in it.

The references cannot apportion the effect. Both shuffle references are biased toward agreement and change which cases are detected, so they bound nothing (Section 3).

The headline exemplar is a weak model. The transformer's 24 minutes comes with case-level AUROC near 0.65, and no one would deploy it at that operating point. The window dependence does not rest on it (Section 4), but the specificity contrast is sharpest there.

Affordance, not comprehension, and one domain. We analyze what a reporting convention makes inferable, not what readers actually infer; measuring that is the next study, and Section 6.3's items are what it would test. The four items are demonstrated here, not validated across settings, and three cohorts in two countries still give one procedure setting and one monitored variable. The seven reports of Table 3 show that the omission occurs, not how often; a systematic survey of reported lead times would.

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Appendix A Methods and complete per-cell results

A.1 Cohorts, models, training and compute

Cohorts and eligibility. Two public datasets supply the three cohorts: VitalDB, and MOVER, whose single release carries both an EPIC-derived and an SIS-derived arm. Both are de-identified. VitalDB v1.0.0 is openly accessible from PhysioNet under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International (CC BY-NC-SA 4.0) license; MOVER is available from the UCI Machine Learning Repository to researchers who sign its data use agreement. Development used MOVER's EPIC arm. The subject-aware split at seed 1004 gave 6,409 training, 1,375 validation and 1,404 test cases out of 9,188, split on unique subject identifiers so that a patient with several operations lands in one fold. The test split holds 1,404 cases, 786 with a sustained event, 1,145 subjects. External evaluation used VitalDB, with 2,804 cases, 1,007 events and 2,730 subjects, and the SIS-derived arm of MOVER, with 2,641 cases and 1,409 events, neither of which contributed to training, calibration or threshold selection. Neither arm of MOVER carries a subject key that resolves against the other, so that statement holds at the level of the released records and cannot be checked at the level of patients. Per-minute eligibility drops the induction prefix, 25 minutes where the record is anchored on anesthesia start, 10 where it is anchored on surgical start, and every minute at or after first onset, so no model is asked to predict an event already in progress. Event cases required five valid pre-onset minutes after that exclusion and controls thirty. Procedure group is never a model input in any family: it separates the cohorts almost perfectly and would be learned as cohort identity.

Training samples, labels and the offset used here. Training uses one sample per case: the 120-minute window ending at that case's reference time, labeled 1 if the case has a sustained event and 0 otherwise. There is no per-minute training target. For an event case the reference time is the first sustained-onset minute shifted back by a lead-time offset, and that offset is **zero for every model reported in this paper**, so the reference time is onset itself. A control has no onset, so it is given a sampled pseudo-onset: its onset ratio, onset divided by case length, is drawn with replacement from the event cases of the same source cohort and multiplied by the control's own case length, then resampled rather than clipped until it falls inside the eligible range, so the control reference times match the event distribution within each source. Event cases with fewer than five eligible pre-onset minutes and controls shorter than thirty are dropped before that sampling. The scored checkpoint is the epoch with the best validation AUROC, and it is the only one used. Evaluation departs from training in the one respect that matters here: every eligible minute of a case is scored independently and inherits its case's label, and `t_to_event` is measured against that same reference time, real onset for an event case and the sampled pseudo-onset for a control.

Features. The same twenty inputs for every family: sixteen per-minute channels, being MAP, systolic and diastolic pressure, their non-invasive counterparts, heart rate, SpO2, end-tidal CO2, temperature, FiO2, MAC, tidal volume, respiratory rate, PEEP and peak inspiratory pressure, and four demographics, being age, sex, height and weight. The families differ in how they consume them. The transformer takes each non-missing cell as a (type, time, value) triplet with demographics as a fixed prefix at time zero, up to 2,048 tokens over a 120-minute window, so missing cells produce no token and are never imputed. The LSTM takes the normalized per-minute window as a value-and-mask pair per channel. Logistic regression, random forest and XGBoost take the last observation carried forward per channel plus the demographics. Per-channel median and interquartile scaling was fitted on the training cases alone and applied unchanged everywhere else.

Architectures and training. Seed 1004 throughout. The transformer is a STraTS-style encoder, a type embedding plus a value MLP plus a time MLP summed per triplet, two encoder layers, model width 128, four heads, feed-forward width 512, dropout 0.1, a learnable classification token and a two-class softmax head; AdamW at learning rate 1.2e-3, weight decay 1e-2 applied to two-dimensional tensors only, gradient clipping at 1.0, a cosine schedule with 5% warmup, batch size 64, cross-entropy, up to 50 epochs with early stopping on validation AUROC at patience 10, under bfloat16 autocast. The LSTM is one layer of width 128 with dropout 0.1, AdamW at 1e-3 with weight decay 1e-2, batch size 64, up to 40 epochs at patience 8. Logistic regression is median imputation and standardization followed by an L2 fit at up to 2,000 iterations. The random forest is 400 trees. XGBoost is 600 rounds at depth 6, learning rate 0.05, row and column subsampling at 0.8, histogram tree method, early stopping after 40 rounds on validation AUC. No hyperparameter search was run over any of these values.

Compute. Every model was trained on one Apple M3 Ultra desktop with 512 GB of unified memory, using PyTorch 2.10 on the Metal backend with bfloat16 autocast; no GPU cluster was used. Wall clock for the five models reported here was 40.3 minutes for the transformer, 14.5 for the LSTM, 3.6 for the random forest, and 34 and 33 seconds for XGBoost and logistic regression. The project trained more than this paper reports: each family was also trained at seven lead-time offsets, of which the transformer arm alone took 3.8 hours. The analyses in this paper add no training and read cached per-minute scores on the same machine. Software was PyTorch 2.10.0, scikit-learn 1.7.1, XGBoost 3.3.0, NumPy 2.2.6, pandas 2.3.2 and SciPy 1.16.1.

Data and code. Both source datasets are public and de-identified. VitalDB v1.0.0 is openly accessible from PhysioNet under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International (CC BY-NC-SA 4.0) license; MOVER is available from the UCI Machine Learning Repository under its data use agreement, one download supplying both arms used here. The Institutional Review Board of Ewha Womans University Mokdong Hospital granted an exemption from review and waived informed consent for this secondary analysis of publicly available de-identified data (EUMC 2026-08-011). This work added no data collection and no patient contact. Analysis code is available at `https://github.com/Jovinus/ioh-leadtime`, archived at `https://doi.org/10.5281/zenodo.23881391`, under the Apache License 2.0. The repository carries the analysis code and the aggregate result files behind the tables and figures, and no patient-level or derived patient-level data of any kind; each dataset is obtained from its own provider under the access and license terms above.

A.2 Lead time by quantile-derived band of observed window, all fifteen cells

Table A1 | Median alert lead time by quantile-derived band of observed window, all fifteen cells. Main text Section 3 reports the ITE Transformer. The gradient is positive in every cell. `Bands` is the number of quantile-derived groups actually realized: where the observed window is censored at the 60-minute alert window the upper quintiles collapse onto a single value, and duplicate edges are dropped rather than bands invented. The gradient is the difference of the highest and lowest band medians, and its interval is the paired cluster bootstrap of that difference: one weight matrix is drawn per cell and reused across bands, and the band edges and the window span are held fixed at their full-sample values so the ratio's denominator is not itself resampled. The interval excludes zero in all fifteen cells. Band medians and gradients are displayed rounded to the minute but computed on unrounded values, so where a band median falls on a half-minute a row's displayed endpoints need not differ by exactly the displayed gradient; logistic regression under VitalDB is the one row where they do not. `Lead min per window min` is the ratio of those two between-band median contrasts and not a per-case slope, so a value slightly above 1.0 carries no contradiction with the per-case constraint that a lead cannot exceed its own window.

Model	Cohort	Bands	Window Q1→last, min	Lead Q1→last, min	Gradient, min (95% CI)	Lead min per window min (95% CI)
ITE Transformer	MOVER (EPIC) test	4	10 → 60	9 → 46	+37 [+35, +41]	0.74 [0.70, 0.82]
ITE Transformer	VitalDB	5	7 → 60	5 → 59	+54 [+51, +55]	1.02 [0.96, 1.04]
ITE Transformer	MOVER (SIS)	4	8 → 60	6 → 46	+40 [+37, +43]	0.77 [0.71, 0.83]
LSTM	MOVER (EPIC) test	4	10 → 60	6 → 44	+38 [+35, +41]	0.77 [0.70, 0.82]
LSTM	VitalDB	5	6 → 60	3 → 34	+31 [+21, +48]	0.57 [0.39, 0.89]
LSTM	MOVER (SIS)	4	9 → 60	5 → 31	+26 [+19, +30]	0.51 [0.37, 0.59]
XGBoost	MOVER (EPIC) test	3	12 → 60	8 → 38	+30 [+26, +34]	0.62 [0.54, 0.71]
XGBoost	VitalDB	3	9 → 60	3 → 30	+27 [+22, +30]	0.53 [0.43, 0.58]
XGBoost	MOVER (SIS)	3	12 → 60	5 → 36	+31 [+28, +33]	0.65 [0.58, 0.69]
Random forest	MOVER (EPIC) test	3	13 → 60	8 → 38	+30 [+26, +34]	0.64 [0.55, 0.72]
Random forest	VitalDB	3	9 → 60	3 → 34	+31 [+29, +34]	0.61 [0.57, 0.67]
Random forest	MOVER (SIS)	3	12 → 60	6 → 40	+34 [+31, +36]	0.71 [0.65, 0.75]
Logistic regression	MOVER (EPIC) test	3	11 → 60	8 → 40	+32 [+28, +35]	0.65 [0.57, 0.71]
Logistic regression	VitalDB	3	10 → 60	3 → 28	+26 [+20, +28]	0.51 [0.40, 0.56]
Logistic regression	MOVER (SIS)	3	12 → 60	5 → 38	+33 [+29, +36]	0.69 [0.61, 0.75]

A.3 Lead time by band, the transformer's three cohorts

Table A2 | Median alert lead time inside quantile-derived bands of observed window, ITE Transformer. ITE Transformer, three held-out cohorts, at the transported 80%-sensitivity operating point. Quintile edges are cut within each cell on detected cases; where the observed window is censored at the 60-minute alert window the upper quintiles collapse and the realized band count is reported rather than assumed. `Observed window, min` is the median observed pre-onset window of that band's cases rather than an edge. Intervals are 95% cluster bootstrap over subjects, or over cases in MOVER (SIS). All fifteen cells are Table A1; the detection rate these bands condition on is Table A3. Section 3 reports the contrast this table carries.

Cohort	Band	n	Observed window, min	Median lead, min (95% CI)	Lead / window (95% CI)
MOVER (EPIC) test	1	139	10	9 [8, 9]	0.88 [0.85, 0.89]
MOVER (EPIC) test	2	111	19	18 [17, 18]	0.94 [0.94, 0.95]
MOVER (EPIC) test	3	121	33	31 [29, 32]	0.97 [0.96, 0.97]
MOVER (EPIC) test	4	240	60	46 [44, 50]	0.98 [0.96, 0.98]
VitalDB	1	143	7	5 [5, 6]	0.83 [0.83, 0.83]
VitalDB	2	140	12	10 [10, 11]	0.91 [0.90, 0.91]
VitalDB	3	132	20	19 [18, 20]	0.94 [0.94, 0.95]
VitalDB	4	134	40	38 [37, 40]	0.97 [0.97, 0.97]
VitalDB	5	135	60	59 [57, 60]	0.98 [0.98, 1.00]
MOVER (SIS)	1	199	8	6 [5, 6]	0.80 [0.80, 0.82]
MOVER (SIS)	2	179	17	14 [13, 15]	0.86 [0.84, 0.91]
MOVER (SIS)	3	186	29	27 [26, 28]	0.94 [0.93, 0.96]
MOVER (SIS)	4	362	60	46 [43, 49]	0.95 [0.93, 0.97]

A.4 Detection rate by band of observed window, all fifteen cells

Table A3 | Detection rate by band of observed pre-onset window, all fifteen cells. Bands are quantiles of the observed window cut over **all** event cases with a positive observed window, not over the detected subset Table A2 and Table A1 cut on, because a within-band detection rate needs every event case of that window length in its denominator. The two banding schemes are therefore not interchangeable; the band counts here are event-case counts rather than detected counts, and the median leads differ from Table A1's, sometimes by several minutes. The detection rate summed over bands reproduces the case-level sensitivity of Table A6 exactly in all fifteen cells, which is the check that the denominators are the same event cases. Intervals are 95% cluster bootstrap.

Model	Cohort	Band	Event cases	Detected	Window median, min	Detection rate (95% CI)	Median lead, min (95% CI)
ITE Transformer	MOVER (EPIC) test	1	159	159	11	1.000 [1.000, 1.000]	9 [8, 10]
ITE Transformer	MOVER (EPIC) test	2	164	164	23	1.000 [1.000, 1.000]	22 [20, 24]
ITE Transformer	MOVER (EPIC) test	3	150	149	44	0.993 [0.979, 1.000]	42 [40, 43]
ITE Transformer	MOVER (EPIC) test	4	313	139	60	0.444 [0.388, 0.498]	54 [33, 58]
ITE Transformer	VitalDB	1	204	204	8	1.000 [1.000, 1.000]	6 [6, 7]
ITE Transformer	VitalDB	2	203	203	18	1.000 [1.000, 1.000]	16 [14, 17]
ITE Transformer	VitalDB	3	600	277	60	0.462 [0.421, 0.500]	44 [42, 46]

Model	Cohort	Band	Event cases	Detected	Window median, min	Detection rate (95% CI)	Median lead, min (95% CI)
ITE Transformer	MOVER (SIS)	1	287	283	10	0.986 [0.971, 0.997]	7 [7, 8]
ITE Transformer	MOVER (SIS)	2	285	281	25	0.986 [0.971, 0.997]	23 [22, 24]
ITE Transformer	MOVER (SIS)	3	837	362	60	0.432 [0.398, 0.465]	46 [43, 49]
LSTM	MOVER (EPIC) test	1	159	148	11	0.931 [0.889, 0.970]	7 [6, 8]
LSTM	MOVER (EPIC) test	2	164	157	23	0.957 [0.925, 0.987]	20 [18, 21]
LSTM	MOVER (EPIC) test	3	150	141	44	0.940 [0.897, 0.974]	39 [37, 41]
LSTM	MOVER (EPIC) test	4	313	188	60	0.601 [0.545, 0.652]	40 [34, 46]
LSTM	VitalDB	1	204	166	8	0.814 [0.760, 0.868]	3 [3, 4]
LSTM	VitalDB	2	203	147	18	0.724 [0.660, 0.783]	10 [9, 12]
LSTM	VitalDB	3	600	182	60	0.303 [0.268, 0.340]	32 [28, 34]
LSTM	MOVER (SIS)	1	287	196	10	0.683 [0.628, 0.738]	6 [5, 6]
LSTM	MOVER (SIS)	2	285	188	25	0.660 [0.604, 0.714]	18 [16, 21]
LSTM	MOVER (SIS)	3	837	393	60	0.470 [0.437, 0.504]	33 [29, 36]
XGBoost	MOVER (EPIC) test	1	159	110	11	0.692 [0.619, 0.760]	7 [5, 9]
XGBoost	MOVER (EPIC) test	2	164	132	23	0.805 [0.743, 0.865]	17 [16, 19]
XGBoost	MOVER (EPIC) test	3	150	130	44	0.867 [0.806, 0.918]	32 [28, 36]
XGBoost	MOVER (EPIC) test	4	313	265	60	0.847 [0.806, 0.885]	40 [34, 44]
XGBoost	VitalDB	1	204	110	8	0.539 [0.471, 0.606]	3 [2, 3]
XGBoost	VitalDB	2	203	132	18	0.650 [0.587, 0.718]	7 [5, 9]
XGBoost	VitalDB	3	600	457	60	0.762 [0.728, 0.794]	26 [23, 30]
XGBoost	MOVER (SIS)	1	287	179	10	0.624 [0.570, 0.678]	5 [4, 5]

Model	Cohort	Band	Event cases	Detected	Window median, min	Detection rate (95% CI)	Median lead, min (95% CI)
XGBoost	MOVER (SIS)	2	285	228	25	0.800 [0.753, 0.844]	14 [13, 16]
XGBoost	MOVER (SIS)	3	837	706	60	0.843 [0.819, 0.866]	35 [33, 37]
Random forest	MOVER (EPIC) test	1	159	107	11	0.673 [0.596, 0.746]	6 [5, 8]
Random forest	MOVER (EPIC) test	2	164	135	23	0.823 [0.762, 0.880]	17 [15, 18]
Random forest	MOVER (EPIC) test	3	150	129	44	0.860 [0.800, 0.914]	30 [27, 34]
Random forest	MOVER (EPIC) test	4	313	280	60	0.895 [0.861, 0.928]	41 [35, 45]
Random forest	VitalDB	1	204	140	8	0.686 [0.626, 0.747]	3 [2, 3]
Random forest	VitalDB	2	203	157	18	0.773 [0.711, 0.831]	7 [6, 10]
Random forest	VitalDB	3	600	512	60	0.853 [0.825, 0.882]	32 [30, 35]
Random forest	MOVER (SIS)	1	287	201	10	0.700 [0.646, 0.752]	5 [5, 6]
Random forest	MOVER (SIS)	2	285	245	25	0.860 [0.819, 0.899]	16 [14, 17]
Random forest	MOVER (SIS)	3	837	750	60	0.896 [0.875, 0.916]	38 [36, 41]
Logistic regression	MOVER (EPIC) test	1	159	119	11	0.748 [0.678, 0.814]	7 [6, 9]
Logistic regression	MOVER (EPIC) test	2	164	138	23	0.841 [0.787, 0.896]	18 [17, 20]
Logistic regression	MOVER (EPIC) test	3	150	134	44	0.893 [0.843, 0.940]	36 [32, 38]
Logistic regression	MOVER (EPIC) test	4	313	267	60	0.853 [0.812, 0.892]	42 [36, 46]
Logistic regression	VitalDB	1	204	71	8	0.348 [0.283, 0.412]	3 [2, 3]
Logistic regression	VitalDB	2	203	95	18	0.468 [0.404, 0.539]	7 [6, 8]
Logistic regression	VitalDB	3	600	298	60	0.497 [0.454, 0.537]	26 [22, 30]
Logistic regression	MOVER (SIS)	1	287	178	10	0.620 [0.566, 0.676]	5 [5, 5]
Logistic regression	MOVER (SIS)	2	285	221	25	0.775 [0.727, 0.821]	14 [12, 16]

Model	Cohort	Band	Event cases	Detected	Window median, min	Detection rate (95% CI)	Median lead, min (95% CI)
Logistic regression	MOVER (SIS)	3	837	714	60	0.853 [0.829, 0.876]	36 [34, 38]

A.5 The relationship in all five families

Figure 1 plots the transformer in three cohorts. The lead-to-window relationship it shows is not particular to that model: in the per-case scatter for MOVER (EPIC) test it is present in every family, with the transformer the limiting case rather than the exception (Figure 2, in the main text). Figure 2 covers that cohort only; the other two are in Table A1, where the gradient is positive with its interval excluding zero in every cell.

A.6 Both references, all fifteen cells

Table A4 | Both references, all fifteen cells. Observed ratio against the uniform-placement reference of Section 3 and both variants of the cross-case shuffle. The uniform reference's expectation is 0.5 by construction; the two shuffle variants differ only in how a donor shorter than its recipient is handled, and both bias the reference upward. Intervals are the 2.5th and 97.5th percentiles of the reference distribution.

Model	Cohort	Observed ratio	Uniform ref.	Shuffle ref., hold	Shuffle ref., length-matched	Observed ρ	Shuffle ρ, hold	Shuffle ρ, length-matched
ITE Transformer	MOVER (EPIC) test	0.944	0.524 [0.491, 0.567]	0.970 [0.967, 0.974]	0.962 [0.951, 0.971]	0.706	0.888 [0.855, 0.920]	0.783 [0.705, 0.856]
ITE Transformer	VitalDB	0.923	0.532 [0.500, 0.573]	0.966 [0.963, 0.967]	0.957 [0.947, 0.969]	0.906	0.941 [0.926, 0.955]	0.933 [0.890, 0.967]
ITE Transformer	MOVER (SIS)	0.899	0.527 [0.500, 0.567]	0.949 [0.941, 0.950]	0.918 [0.900, 0.938]	0.801	0.877 [0.855, 0.895]	0.775 [0.713, 0.827]
LSTM	MOVER (EPIC) test	0.851	0.522 [0.483, 0.563]	0.913 [0.900, 0.923]	0.866 [0.833, 0.893]	0.650	0.825 [0.784, 0.860]	0.673 [0.596, 0.743]
LSTM	VitalDB	0.619	0.534 [0.500, 0.583]	0.870 [0.850, 0.883]	0.752 [0.667, 0.833]	0.759	0.845 [0.798, 0.886]	0.705 [0.593, 0.804]
LSTM	MOVER (SIS)	0.717	0.524 [0.500, 0.562]	0.873 [0.857, 0.887]	0.700 [0.633, 0.767]	0.501	0.682 [0.636, 0.727]	0.455 [0.368, 0.547]
XGBoost	MOVER (EPIC) test	0.719	0.519 [0.483, 0.558]	0.842 [0.814, 0.867]	0.760 [0.705, 0.800]	0.558	0.701 [0.651, 0.751]	0.581 [0.503, 0.649]
XGBoost	VitalDB	0.467	0.521 [0.483, 0.556]	0.723 [0.667, 0.768]	0.591 [0.517, 0.652]	0.599	0.636 [0.569, 0.698]	0.493 [0.400, 0.580]
XGBoost	MOVER (SIS)	0.600	0.518 [0.500, 0.550]	0.779 [0.750, 0.800]	0.681 [0.633, 0.722]	0.573	0.632 [0.590, 0.673]	0.488 [0.429, 0.541]
Random forest	MOVER (EPIC) test	0.714	0.519 [0.483, 0.556]	0.839 [0.816, 0.867]	0.777 [0.733, 0.817]	0.584	0.714 [0.665, 0.764]	0.599 [0.531, 0.667]
Random forest	VitalDB	0.533	0.523 [0.500, 0.556]	0.764 [0.721, 0.800]	0.685 [0.625, 0.733]	0.651	0.678 [0.622, 0.729]	0.555 [0.479, 0.627]
Random forest	MOVER (SIS)	0.667	0.518 [0.500, 0.550]	0.811 [0.783, 0.833]	0.741 [0.700, 0.783]	0.618	0.659 [0.625, 0.692]	0.536 [0.486, 0.589]
Logistic regression	MOVER (EPIC) test	0.781	0.520 [0.483, 0.557]	0.872 [0.849, 0.898]	0.802 [0.767, 0.842]	0.597	0.739 [0.694, 0.779]	0.612 [0.544, 0.676]
Logistic regression	VitalDB	0.455	0.521 [0.471, 0.567]	0.721 [0.667, 0.781]	0.577 [0.500, 0.650]	0.607	0.642 [0.549, 0.720]	0.486 [0.374, 0.584]
Logistic regression	MOVER (SIS)	0.618	0.518 [0.500, 0.550]	0.782 [0.750, 0.800]	0.734 [0.692, 0.767]	0.597	0.644 [0.607, 0.682]	0.548 [0.495, 0.595]

Three cells in this table are the ones Section 3 names as not exceeding the uniform reference, all of them VitalDB. The observed ratio sits below it for XGBoost by 0.054 and for logistic regression by 0.067, and random forest is indistinguishable from it (+0.011, $P = 0.824$). For the transformer the excess over the uniform reference is +0.372 to +0.421. Under the length-matched variant the transformer's ratio reference spans 0.92 to 0.96. Every contrast quoted here is computed from the unrounded values rather than from the rounded columns above, so recomputing one from the table can differ in the third decimal.

A.7 Score against elapsed monitoring time

Section 4 reads the first ten monitored minutes of every case, and reports that the ITE Transformer clears its own threshold there in every cohort. The margins for the others are wide. Over those ten minutes the highest median score reached in either group is 0.69 against a threshold of 0.78 for logistic regression in MOVER (EPIC), 0.65 against a threshold of 0.91 for XGBoost, and 0.54 against a threshold of 0.79 for random forest. The LSTM is the one exception: in MOVER (EPIC) it reaches 0.99 in event and 0.99 in control cases against a threshold of 0.97, clearing it in both groups, and it clears it in neither external cohort. This is that view for all fifteen cells, on the elapsed axis rather than the event-aligned one of Figure A2.

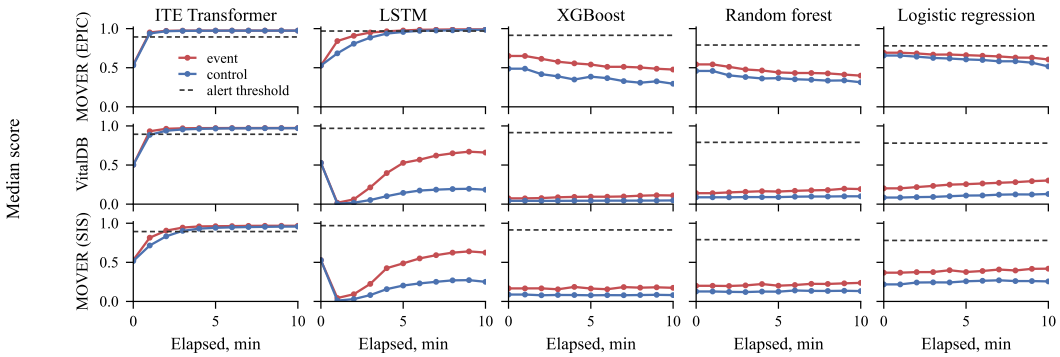


Figure A1: **Median score against elapsed monitoring time, with each family's own alert threshold.** Rows are cohorts, columns are model families. Elapsed time is measured from the case's first scored minute. The dashed rule in each panel is that family's transported raw-score threshold at the 80%-sensitivity target, transferred unchanged from the MOVER (EPIC) validation split. Only the ITE Transformer's early rise clears its own threshold in every cohort, and it does so in event and control cases together; the LSTM does the same in MOVER (EPIC) alone. The elevation ends around minute 20, past the right edge of this axis. This axis is elapsed time; the event-aligned view is Figure A2.

A.8 Event and control score on the event-aligned axis

The shape a reader expects of a continuously read score, rising in the cases that go on to have an event and much flatter in the rest, is **present in all fifteen cells** (Figure A2). Three things about it are worth reporting, and none is visible in a trend statistic.

When the separation arrives runs opposite to near-onset discrimination. For the three tabular families the event and control means are already distinguishable 60 minutes before onset in every cohort, the gap at that range running +0.049 to +0.109 with each pointwise interval excluding zero. For the transformer the gap at 60 minutes **covers zero in all three cohorts** (-0.023 [-0.080, 0.035], -0.014 [-0.057, 0.030] and -0.006 [-0.043, 0.032]), and first excludes it at 35, 20 and 30 minutes. The LSTM sits between the two. The family whose scores separate latest is the one with the highest discrimination near onset.

The event mean rises toward onset, and for the transformer the control mean falls by as much. Over the hour the transformer's event mean rises +0.239, +0.266 and +0.269, which is the tendency a reader expects and it is substantial, while its control mean falls -0.252, -0.263 and -0.196, so about half of the widening is carried by the controls. The LSTM's control mean also falls, by 0.059 to 0.185. For the tabular families the control mean moves between -0.070 and +0.021, rising in several cells, and the widening is almost entirely event-side. So "the score rises in patients who are about to have an event" holds in every cell here; what it leaves out is that for the transformer an equal share of the widening is the control mean coming down, and an event-only trend statistic cannot separate the two.

The split follows the representation more than the architecture. The control side contributes most where the window is read as a sequence and least where the model reads one feature vector per

scored minute, carrying each channel's last observation forward, so it sees a level and has little shape to drift with. The ordering is not clean: the tabular families' control means fall in MOVER (EPIC) as well and only flatten in the two external cohorts. Representation and family are not crossed in this design, so this is a reading the data supports rather than a hypothesis it tests. It does say where to look: **the control-side contribution is a thing to measure in models given freer access to the input series**, and reporting the two sides of a widening gap separately is what makes it visible.

Three things this figure is not. The control series is aligned on the sampled pseudo-onset of Section 2 rather than on an event, so its fall is behavior over the last hour of a comparable window and not an approach to anything. Figure A1 plots medians and this figure means, and the two are not interchangeable here. And the two figures use different alignments: Figure A1's axis is elapsed time from the case's first scored minute and covers the first ten minutes of the record, where the transformer has just settled above its own 0.894 threshold, whereas the bins here sit far later in the case, where its event mean is 0.383. **The two are not in conflict, and the length of the plateau is what reconciles them.** The median score holds at or above the transported threshold through elapsed minute 24, 21 and 17 in event cases and 23, 20 and 16 in controls, then falls to 0.03 to 0.31 by minute 30. Figure A1 covers the first ten of those minutes and stops inside the plateau; every bin in this figure falls after it in all but the shortest records.

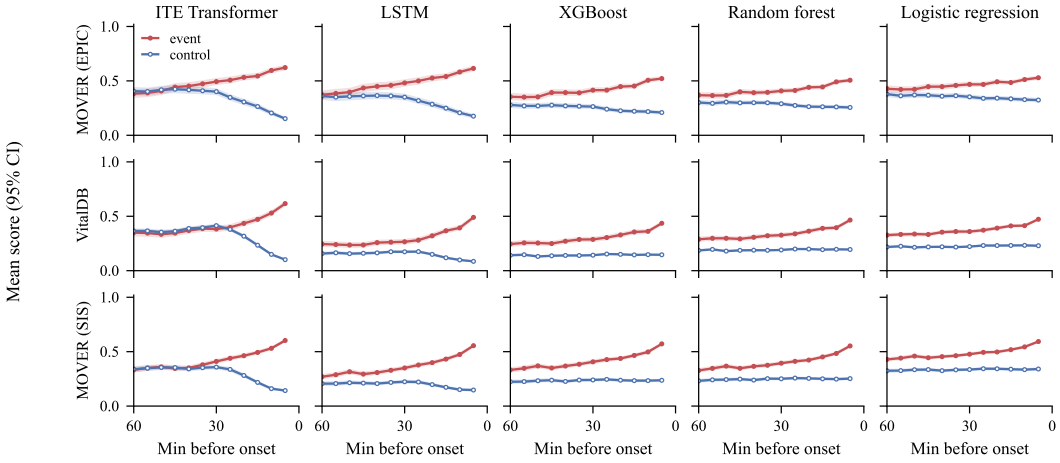


Figure A2: **Mean score in event and control cases against minutes before onset.** Rows are cohorts, columns are model families, the same grid as Figure A1, on the event-aligned axis instead of the elapsed one. Event cases solid and filled, control cases dashed and open; bands are 95% cluster bootstrap. Bins are the 13 centers at five-minute spacing from 60 minutes before onset; the bin at onset is empty by construction, because no minute is strictly before its own reference time, so the series ends at 5 minutes. Control cases are aligned on a sampled pseudo-onset.

A.9 Path decomposition

Table A5 | Path decomposition over the window 60 to 5 minutes before onset. Event episodes have observed scores at both window edges. Directedness is net displacement divided by total distance traveled, so 1 is a monotone climb and 0 is a round trip. Path per net is total distance divided by the absolute net change. Episodes observed at every bin center in the window; median 11 steps per episode. Intervals are 95% cluster bootstrap. Only this window is reported: over 60 to 10 minutes the transformer's net displacement is indistinguishable from zero in two cohorts, and the ratio is then a division by approximately nothing.

Cohort	Model	Episodes	Directedness (95% CI)	Path per unit net
MOVER (EPIC) test	ITE Transformer	313	0.025 [0.002, 0.050]	40.7
MOVER (EPIC) test	LSTM	313	0.043 [0.024, 0.057]	25.9
MOVER (EPIC) test	XGBoost	313	0.049 [0.033, 0.070]	20.0
MOVER (EPIC) test	Random forest	313	0.067 [0.038, 0.093]	16.3
MOVER (EPIC) test	Logistic regression	313	0.061 [0.044, 0.089]	17.6
VitalDB	ITE Transformer	433	0.021 [0.005, 0.038]	146.6
VitalDB	LSTM	433	0.059 [0.035, 0.070]	28.2
VitalDB	XGBoost	433	0.069 [0.049, 0.099]	17.9
VitalDB	Random forest	433	0.075 [0.053, 0.094]	14.4
VitalDB	Logistic regression	433	0.088 [0.063, 0.108]	12.6
MOVER (SIS)	ITE Transformer	670	0.089 [0.066, 0.122]	13.1
MOVER (SIS)	LSTM	670	0.142 [0.118, 0.183]	7.9
MOVER (SIS)	XGBoost	670	0.144 [0.120, 0.167]	7.5
MOVER (SIS)	Random forest	670	0.151 [0.132, 0.168]	7.0
MOVER (SIS)	Logistic regression	670	0.134 [0.119, 0.154]	7.9

Directedness is net displacement divided by total distance traveled: over the window from 60 to 5 minutes before onset it is **+0.021 to +0.151 across all fifteen cells**, each pointwise interval excluding zero, so no family approaches a directed climb. **Directedness describes path shape and does not establish when or whether a particular threshold is touched.** Three further things belong here. The ratio of the median path length to the absolute median net displacement is 7.0 to 146.6 across the fifteen cells; because that ratio becomes unstable as net displacement approaches zero, directedness is the primary summary and both quantities describe path shape rather than prove a crossing mechanism. The transformer sits at the volatile end of both (directedness +0.021 to +0.089, path per unit net displacement 13.1 to 146.6) and logistic regression at the least volatile (+0.061 to +0.134, 7.9 to 17.6). And averaged over cases the score still behaves as a reader expects, rising where an event follows and staying much flatter where none does, but the transformer's event and control means are indistinguishable an hour before onset where the tabular families' are already apart (Figure A2). The property is present under this task and scoring policy and the transformer is its limiting case; the design does not isolate architecture from representation or training.

A.10 Case-level operating point, complete detail

Table A6 | Case-level operating point, all fifteen cells. This is the operating point behind every reported lead time in the paper: every family, every cohort and every interval, and it is where the cross-combination claim of Section 4 is checked. Sensitivity and specificity are case-level as defined in Section 2, NNE is alerting cases per detected event and equals 1/PPV, and `Min. above thr.` is the median share of an event case's monitored minutes at or above the transported threshold. Cohort composition, prevalence, PPV in excess of prevalence, the control-side minute share and the share of event cases already alerting by the first elapsed minute are Table A7. Intervals are 95% cluster bootstrap over subjects, or over cases in MOVER (SIS). Section 4's cross-combination rank correlations are computed from the unrounded per-cell values rather than from the rounded columns here: recomputing the PPV-lift correlation from PPV and prevalence each rounded to three decimals gives -0.727 instead of -0.719, and the specificity correlation is unchanged at -0.803 either way.

Model	Cohort	Sensitivity (95% CI)	Specificity (95% CI)	PPV (95% CI)	NNE (95% CI)	Min. above thr., events (95% CI)	Median lead, min (95% CI)
ITE Transformer	MOVER (EPIC) test	0.777 [0.746, 0.806]	0.406 [0.368, 0.443]	0.625 [0.594, 0.656]	1.60 [1.53, 1.68]	0.523 [0.444, 0.575]	24 [21, 27]
ITE Transformer	VitalDB	0.679 [0.651, 0.707]	0.379 [0.358, 0.403]	0.380 [0.357, 0.402]	2.63 [2.49, 2.80]	0.444 [0.389, 0.500]	17 [15, 19]
ITE Transformer	MOVER (SIS)	0.657 [0.632, 0.682]	0.467 [0.441, 0.494]	0.585 [0.561, 0.609]	1.71 [1.64, 1.78]	0.200 [0.150, 0.238]	22 [20, 24]
LSTM	MOVER (EPIC) test	0.807 [0.779, 0.834]	0.474 [0.433, 0.513]	0.661 [0.631, 0.690]	1.51 [1.45, 1.58]	0.267 [0.209, 0.308]	23 [20, 25]
LSTM	VitalDB	0.492 [0.460, 0.523]	0.812 [0.795, 0.830]	0.595 [0.563, 0.629]	1.68 [1.59, 1.78]	0.000 [0.000, 0.017]	9 [8, 10]
LSTM	MOVER (SIS)	0.551 [0.526, 0.577]	0.709 [0.683, 0.735]	0.685 [0.657, 0.712]	1.46 [1.40, 1.52]	0.017 [0.017, 0.020]	15 [14, 18]
XGBoost	MOVER (EPIC) test	0.810 [0.782, 0.835]	0.608 [0.570, 0.647]	0.725 [0.696, 0.754]	1.38 [1.33, 1.44]	0.117 [0.100, 0.133]	21 [18, 23]
XGBoost	VitalDB	0.694 [0.665, 0.721]	0.754 [0.735, 0.774]	0.613 [0.584, 0.640]	1.63 [1.56, 1.71]	0.067 [0.050, 0.067]	13 [12, 16]
XGBoost	MOVER (SIS)	0.790 [0.768, 0.810]	0.638 [0.610, 0.666]	0.714 [0.691, 0.736]	1.40 [1.36, 1.45]	0.100 [0.100, 0.117]	19 [18, 22]
Random forest	MOVER (EPIC) test	0.828 [0.801, 0.854]	0.555 [0.515, 0.597]	0.703 [0.674, 0.732]	1.42 [1.37, 1.48]	0.150 [0.133, 0.167]	22 [20, 24]
Random forest	VitalDB	0.803 [0.778, 0.827]	0.674 [0.651, 0.695]	0.580 [0.553, 0.604]	1.72 [1.65, 1.81]	0.133 [0.118, 0.150]	15 [12, 18]
Random forest	MOVER (SIS)	0.849 [0.830, 0.867]	0.559 [0.532, 0.586]	0.688 [0.666, 0.710]	1.45 [1.41, 1.50]	0.172 [0.167, 0.183]	22 [20, 25]
Logistic regression	MOVER (EPIC) test	0.837 [0.810, 0.864]	0.518 [0.477, 0.557]	0.688 [0.658, 0.719]	1.45 [1.39, 1.52]	0.139 [0.114, 0.162]	23 [21, 26]
Logistic regression	VitalDB	0.461 [0.431, 0.492]	0.825 [0.808, 0.842]	0.596 [0.565, 0.631]	1.68 [1.59, 1.77]	0.000 [0.000, 0.000]	14 [10, 16]
Logistic regression	MOVER (SIS)	0.790 [0.769, 0.812]	0.585 [0.559, 0.611]	0.685 [0.662, 0.708]	1.46 [1.41, 1.51]	0.117 [0.100, 0.130]	21 [19, 23]

A.11 Cohort composition and the control side

Table A7 | Cohort composition, PPV against prevalence, and the control side. `PPV − prevalence` is the lift over the base rate of the cohort the PPV is measured in, because raw PPV is not comparable across cohorts whose prevalence runs from 0.359 to 0.560. `Alerted by elapsed min 1` is one minus the Kaplan-Meier survival of time to first alert at the first elapsed monitoring minute, censored at the end of each case's own window; it is the censoring-aware form of Section 4's settling. It is **not** the prevalent share of Section 5, which uses a different numerator and a different denominator; Appendix A.13 sets the two estimands out side by side. Intervals are 95% cluster bootstrap.

Model	Cohort	Event cases	Control cases	Prevalence	PPV − prevalence	Min. above thr., controls (95% CI)	Alerted by elapsed min 1 (95% CI)
ITE Transformer	MOVER (EPIC) test	786	618	0.560	+0.065	0.150 [0.050, 0.233]	0.532 [0.497, 0.565]
ITE Transformer	VitalDB	1007	1797	0.359	+0.021	0.183 [0.163, 0.233]	0.460 [0.428, 0.491]
ITE Transformer	MOVER (SIS)	1409	1232	0.534	+0.051	0.025 [0.017, 0.047]	0.308 [0.285, 0.332]
LSTM	MOVER (EPIC) test	786	618	0.560	+0.101	0.017 [0.000, 0.027]	0.228 [0.198, 0.257]
LSTM	VitalDB	1007	1797	0.359	+0.236	0.000 [0.000, 0.000]	0.087 [0.071, 0.105]
LSTM	MOVER (SIS)	1409	1232	0.534	+0.151	0.000 [0.000, 0.000]	0.138 [0.120, 0.157]
XGBoost	MOVER (EPIC) test	786	618	0.560	+0.165	0.000 [0.000, 0.000]	0.188 [0.161, 0.215]
XGBoost	VitalDB	1007	1797	0.359	+0.253	0.000 [0.000, 0.000]	0.062 [0.047, 0.077]
XGBoost	MOVER (SIS)	1409	1232	0.534	+0.180	0.000 [0.000, 0.000]	0.119 [0.101, 0.136]
Random forest	MOVER (EPIC) test	786	618	0.560	+0.143	0.000 [0.000, 0.000]	0.183 [0.157, 0.212]
Random forest	VitalDB	1007	1797	0.359	+0.221	0.000 [0.000, 0.000]	0.101 [0.082, 0.121]
Random forest	MOVER (SIS)	1409	1232	0.534	+0.154	0.000 [0.000, 0.000]	0.166 [0.147, 0.186]
Logistic regression	MOVER (EPIC) test	786	618	0.560	+0.128	0.000 [0.000, 0.017]	0.226 [0.198, 0.255]
Logistic regression	VitalDB	1007	1797	0.359	+0.237	0.000 [0.000, 0.000]	0.036 [0.025, 0.048]
Logistic regression	MOVER (SIS)	1409	1232	0.534	+0.152	0.000 [0.000, 0.000]	0.121 [0.104, 0.138]

A.12 Departure timing: searched for, not found

We asked whether the score departs from its own baseline before mean arterial pressure departs from its, within an episode, under a pre-specified rule: baseline is the median of the episode's first ten eligible minutes, the robust scale is 1.4826 $\times$ the median absolute deviation over the same minutes, and a departure is the first minute beginning a run of at least three consecutive minutes more than 1.5 scale units in the deteriorating direction.

Score departure does not precede MAP departure. For XGBoost the share of episodes in which the score departs first is significantly *below* one half in all three cohorts. For the transformer the score does not depart at all in most episodes, because its baseline sits at the top of its range: a median baseline of 0.94 to 0.97 leaves median headroom of 0.005 to 0.031, which is less than the bar its own rule sets in all three cohorts, so there is not enough room above baseline for a departure to register. Both facts are reported so the negative cannot be read as an unfair rule. Sensitivity analyses at a 2.0 scale threshold, a five-minute run length, and a search capped at the 60-minute window all leave the direction unchanged.

The question was pre-specified and the answer constrains Section 4: whatever produces the reported lead, it is not the score moving before the monitored variable does.

A.13 Alert composition

Section 5 reports the prevalent share as a single number per cell. For the transformer that share is 8.7, 9.6 and 9.0% in the three cohorts, and removing those cases moves its median lead from 24, 17 and 22 minutes to 21, 15 and 19. Splitting detected event cases by when their first crossing fell separates alerts that were already active at the first eligible minute from those raised while the score was still settling and those raised later, which is the disclosure Section 6.3's first item asks for.

That share and Table A7's `Alerted by elapsed min 1` are different quantities rather than two estimates of one. The prevalent share counts detected event cases whose first crossing falls at elapsed minute zero, the first minute of the case's own alert window, over the detected event cases of that cell: 53 of 611, 66 of 684 and 83 of 926 for the transformer. Table A7's column counts event cases whose first crossing falls at elapsed minute zero **or** minute one, over every event case with a positive observed window, detected or not, as one minus the Kaplan-Meier survival of time to first alert with each case censored at the end of its own window: 418 of 786, 463 of 1,007 and 434 of 1,409. Numerator, denominator and conditioning set all differ, and the extra minute is the larger of the differences: minute one alone adds 365, 397 and 351 cases, which is Section 4's settling read on the elapsed axis and not a second estimate of the same immediate-alert probability. No case is censored before elapsed minute one in any of the three cohorts, so the survival estimate at that mark equals the crude share.

A.14 Replacement estimands, all horizons

Table A8 | Fixed-horizon detection rate at all seven horizons, and time to first alert. The fixed-horizon detection rate is the share of event cases observed for at least the stated number of minutes before onset whose first alert came at least that many minutes before onset. Point estimates are shown here; each is a share of the cases in its own column, and its 95% cluster bootstrap interval is computed exactly as elsewhere in this appendix. Section 5 quotes the range across cells at the 10- and 30-minute horizons. `n observed` is the denominator at the first and last horizon, and its fall is the reason a single median is not comparable across cohorts. The 60-minute column is the share of fully observed cases that alerted at the very first opportunity.

Model	Cohort	5 min	10 min	15 min	20 min	30 min	45 min	60 min	n observed ≥5 → ≥60
ITE Transformer	MOVER (EPIC) test	0.733	0.690	0.646	0.605	0.512	0.339	0.170	786 → 311
ITE Transformer	VitalDB	0.638	0.560	0.509	0.463	0.390	0.261	0.154	1007 → 429
ITE Transformer	MOVER (SIS)	0.594	0.547	0.504	0.469	0.377	0.247	0.125	1409 → 665
LSTM	MOVER (EPIC) test	0.721	0.663	0.635	0.597	0.490	0.331	0.109	786 → 311
LSTM	VitalDB	0.343	0.278	0.265	0.223	0.173	0.102	0.030	1007 → 429
LSTM	MOVER (SIS)	0.453	0.392	0.346	0.319	0.253	0.167	0.047	1409 → 665
XGBoost	MOVER (EPIC) test	0.688	0.651	0.615	0.575	0.484	0.367	0.119	786 → 311
XGBoost	VitalDB	0.502	0.470	0.453	0.425	0.366	0.237	0.054	1007 → 429
XGBoost	MOVER (SIS)	0.656	0.597	0.565	0.520	0.460	0.326	0.080	1409 → 665
Random forest	MOVER (EPIC) test	0.709	0.674	0.638	0.611	0.506	0.393	0.138	786 → 311
Random forest	VitalDB	0.589	0.559	0.559	0.530	0.478	0.347	0.103	1007 → 429
Random forest	MOVER (SIS)	0.736	0.674	0.640	0.600	0.526	0.403	0.114	1409 → 665
Logistic regression	MOVER (EPIC) test	0.726	0.692	0.658	0.630	0.547	0.401	0.122	786 → 311
Logistic regression	VitalDB	0.338	0.307	0.303	0.274	0.233	0.137	0.026	1007 → 429
Logistic regression	MOVER (SIS)	0.664	0.611	0.573	0.546	0.474	0.353	0.093	1409 → 665

A.15 The full alert window as a constant

Table A9 | Event cases observed for the whole 60-minute alert window. Restricting to these cases makes the observation window a constant rather than a covariate, which is the full-window restriction of Section 5. `n` is the number of event cases observed for the whole window and `Detected` the share of them that alert at all. The censoring is not independent of the event, because for an event case a short observable run-up is an early-onset event rather than missing data, so this describes what the rule did under the observation these cohorts supplied. Intervals are 95% cluster bootstrap. For the transformer in MOVER (EPIC) the median lead here differs from Table A3's top band by seven minutes although the two populations differ by two cases, because the detected full-window leads leave a gap between 47 and 54 minutes: 53 of 137 sit at the 60-minute ceiling and the lower tail is sparse, so a one-rank shift in the median crosses the gap.

Model	Cohort	n	Detected	Median lead, min (95% CI)
ITE Transformer	MOVER (EPIC) test	311	0.441	47 [33, 59]
ITE Transformer	VitalDB	429	0.247	60 [60, 60]
ITE Transformer	MOVER (SIS)	665	0.304	57 [51, 59]
LSTM	MOVER (EPIC) test	311	0.598	40 [34, 46]
LSTM	VitalDB	429	0.200	33 [21, 45]
LSTM	MOVER (SIS)	665	0.433	28 [22, 33]
XGBoost	MOVER (EPIC) test	311	0.846	40 [34, 44]
XGBoost	VitalDB	429	0.762	33 [30, 36]
XGBoost	MOVER (SIS)	665	0.854	37 [34, 41]
Random forest	MOVER (EPIC) test	311	0.894	41 [35, 45]
Random forest	VitalDB	429	0.851	40 [36, 44]
Random forest	MOVER (SIS)	665	0.910	42 [39, 45]
Logistic regression	MOVER (EPIC) test	311	0.852	42 [36, 45]
Logistic regression	VitalDB	429	0.490	31 [26, 37]
Logistic regression	MOVER (SIS)	665	0.857	40 [37, 43]

A.16 Risk-set composition, and what fixing the cohort does

The population trajectory scores every case that has a minute in the bin, so its denominator shrinks and its composition shifts as the bin moves away from onset. For an event case the observable run-up *is* its time to onset, so a short run-up is not missing data. It is an early-onset event, and the far bins keep only the long run-ups. That is the same selection that drives Section 3.

Fixing the cohort moves the trajectory in both directions (Figure A3). Recomputing the same bins on the cases observable back to 60 minutes raises the far end and lowers the near end, and the two curves cross at 16.4, 12.6 and 14.9 minutes before onset. The first bin reaching AUROC 0.70 moves from 15 minutes in all three cohorts to 20, 35 and 20. Neither curve is the true one: which denominator is used is a reporting choice, it changes the horizon a reader would quote by up to twenty minutes, and it is not currently disclosed. That is the whole of Section 6.3's third item.

Short and long run-ups are different operations, which is why Section 5 calls this censoring non-independent. Splitting event cases at a 40-minute observable run-up, the long group's operations are longer in every cohort, by a standardized mean difference of 0.394, 0.515 and 0.428, the largest imbalance on the list in all three; procedure group and ASA class are also imbalanced beyond 0.1 in each. So a case with a short run-up is not a briefly watched version of a case with a long one.

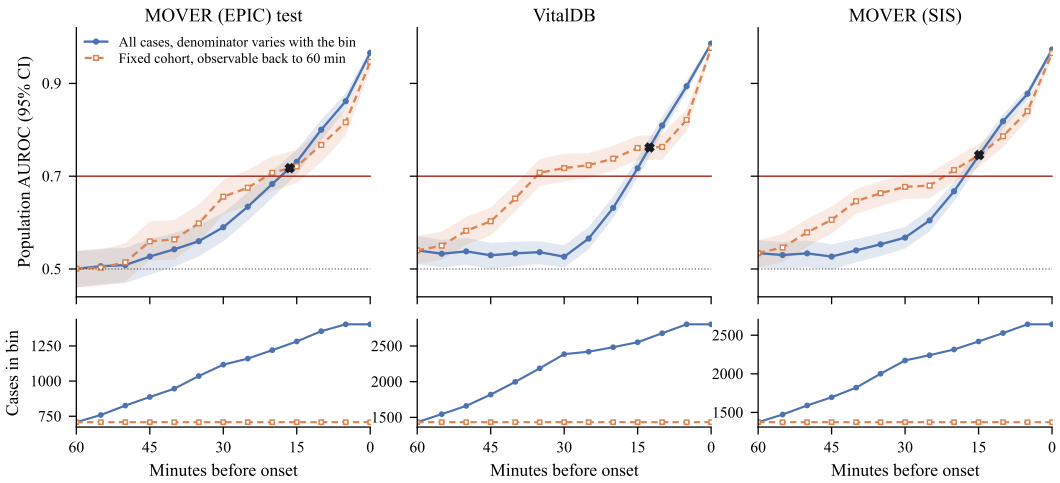


Figure A3: **Population AUROC with its own risk set, event-aligned.** ITE Transformer, 2-minute bins from 60 minutes before onset to onset, in three held-out cohorts. Solid and filled is every case with a minute in the bin, the convention in the published literature, whose denominator varies; dashed and open is the cohort fixed to cases observable back to 60 minutes. Bands are 95% cluster bootstrap over subjects, or over cases in MOVER (SIS). The lower row is the number of cases contributing to each bin, on its own scale per cohort. The dotted line marks 0.5, the solid rule 0.70, and the cross marks where the two curves meet.

A.17 The reporting items that came close

Table A10 | The TRIPOD+AI items that came close and missed. Main text Section 6.1 discusses items 12e, 15, 23a, 20a, 20b and 14; this table records the remaining items, together with the detail behind those the main text discusses. The item numbering is that of the TRIPOD+AI statement [Collins et al., 2024]. This is an author-completed self-assessment of one study, not a validity assessment of the standard.

Item	What it asked for	What it did not ask for
12f and 12g, recalibration and how evaluation predictions were calculated	Any recalibration arising from evaluation, and how evaluation predictions were computed: calibration and thresholds fitted on the development validation split and transported unchanged, which is reported	Anything about what that disciplined choice implies; it is the exact setting in which Section 3 holds, so doing the right thing here does not help
20a and 20b, participant flow and characteristics	Participant flow and characteristics *including follow-up time*, answered here with surgery duration and first-onset time	The distribution of the window the alert rule actually searches — the run-up after the induction exclusion, capped at the 60-minute alert window — or any pairing of it with the timing statistic it bounds. This is the closest the standard comes: it asks for the raw material as a cohort descriptor, and never connects it to the timing metric it constrains
21, event counts	The number of participants and outcome events *in each analysis*, which are reported	The same counts broken out by band of available window. The item's "in each analysis" clause reaches development, tuning and evaluation, not the window strata of a timing statistic
8c, blinding of outcome assessment	A blinding statement; answered "not applicable" because the outcome is computed from a monitored channel that is also a predictor	The closest the checklist comes to Section 4, and still about blinding rather than about what a timing metric measures
15, model output	Details and rationale for classification, and how any threshold was identified; answered with the transported threshold itself	Joint presentation or interpretation of a lead-time summary with the realized operating-point metrics that threshold defines, which is where the artifact lives (Table A6)

A.18 Multiplicity

The two references of Section 3 form one Benjamini–Hochberg family of 75 formal tests, adjusted within that family, of which 57 are significant at $q < 0.05$. The stratification of Section 3 produces intervals rather than tests and does not enter the family. Every bootstrap interval in this paper, including the fifteen window-gradient intervals of Table A1, is a **pointwise descriptive interval**: no family-wise coverage claim is made across the fifteen contrasts, and "each interval excludes zero" is fifteen separate statements rather than one simultaneous one.

A.19 Seeds and reproducibility

Cluster bootstraps use 2,000 replicates, with the clustering unit the subject for two cohorts and the case for MOVER (SIS), whose released subject key is a copy of the case identifier. The two references of Section 3 draw 1,000 permutations per cell. The cross-combination rank correlations of Section 4 are reported as **point estimates with no accompanying test**. A permutation of the pairing was computed while this analysis was being developed, and is not reported: its unit is a cell, and cells sharing a cohort or a model family are not exchangeable, so the resulting P value would overstate what fifteen dependent combinations can establish. Section 4 gives the untied-pair counts and the within-cohort correlations in its place. Model training and the reference-time construction are fixed at a single seed, reported in Appendix A.1.

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